

Titanic Survival Intelligence Suite - Master README

Titanic Survival Intelligence Suite

An enterprise-grade full-stack machine learning web application that predicts Titanic passenger survival using historical demographics and displays interactive analytics, Explainable AI (XAI) SHAP values, batch CSV analysis, and analytical reports.

Platform Features

1. **Dashboard Analytics**: Real-time widgets displaying system usage, active sessions, and classification metrics alongside Recharts graphs (gender, age groups, class volumes).

2. **Infer**
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Project Structure

|---- backend/

| |---- da

Local Development Setup

1. Backend Server Setup

Ensure Python 3.11+ is installed.

cd backe

```
venv\Scripts\pip install -r requirements.txt
```

```
source venv/bin/activate && pip install -r requirements.txt
```

On Li

Run server

```
venv\Scripts\python main.py
```

*Note: On

2. Frontend React Setup

Ensure Node.js is installed.

cd fronte

Production Deployments

Refer [\[deployment.md\]\(file:///C:/Users/SRI%20HARSHA/.gemini/antigravity-ide/brain/369dee50-83e9-4593-9633-58e0b3cf87d9/deployment.md\)](#) to the [guide](#) for deploying the backend on **Render** and the frontend on **Vercel**.